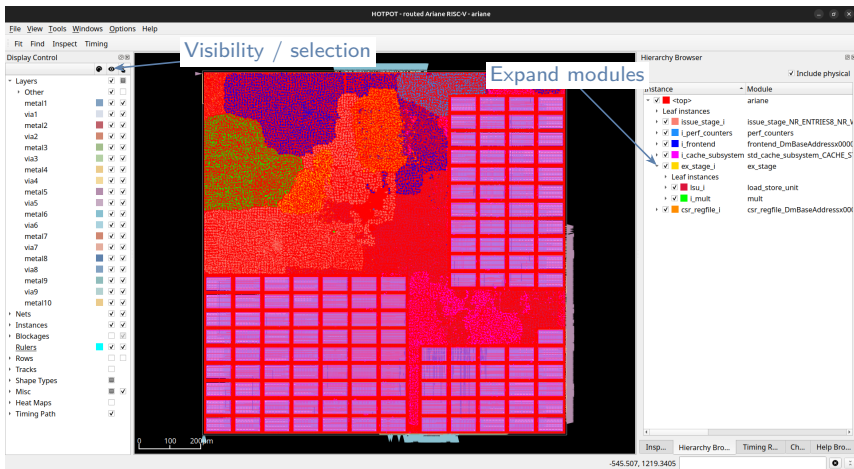
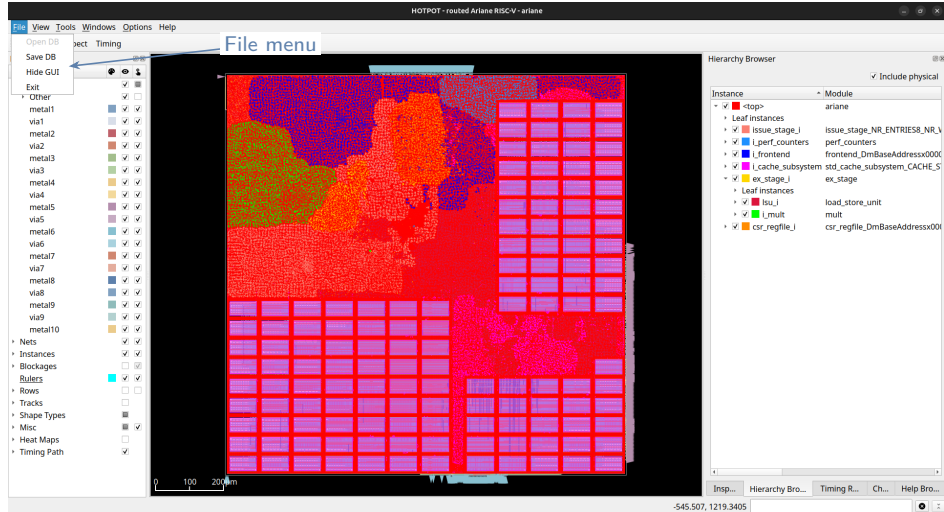


Explore Ariane: a public RISC-V core



133 memory macros · routed design with fillers · Nangate45

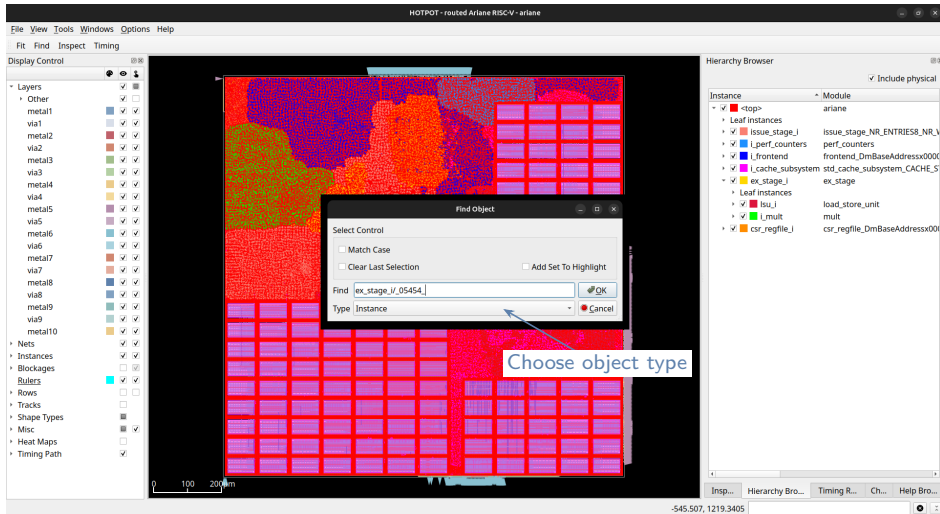
File: open and save databases



Windows: choose the panels



Find an instance by name



Instance: master, placement, terminals

The screenshot displays the HOTPOT - routed Ariane RISC-V - ariane PCB layout editor. The main workspace shows a complex PCB layout with various components and traces. A specific instance is highlighted with a yellow border, labeled "Selected instance". The "Library cell" label points to the instance's library entry, and the "Instance terminals" label points to the instance's terminals.

The left sidebar shows the "Display Control" panel with a list of layers and their visibility status. The "Layers" list includes:

- Other
- metal1
- via1
- metal2
- via2
- metal3
- via3
- metal4
- via4
- metal5
- via5
- metal6
- via6
- metal7
- via7
- metal8
- via8
- metal9
- via9
- metal10
- Nets
- Instances
- Blockages
- Rulers
- Rows
- Tracks
- Shape Types
- Misc
- Heat Maps
- Timing Path

The right sidebar shows the "Inspector" panel with the following properties:

Name	Value
Type	Inst
Name	ex_stage_i/_05454_
Block	ariane
Module	ex_stage_i
Master	INVD2_X1
Description	Multi-Input combinational cell
Placement status	PLACED
Source type	NONE
Dont Touch	False
Orientation	MX
X	394.25 μ m
Y	774.2 μ m
Terminals	5 items
A1	fu_data_o[199]
A2	ex_stage_i/_00031_
ZN	ex_stage_i/_00032_
VDD	VDD
VSS	VSS
Timing/Power	ex_stage_i/_05454_
BBox	(394.25, 774.2), (394.82, 775.6)
BBox Width, Height	(0.57, 1.4)

The bottom status bar shows "Added 1" and "400.7315, 777.299".

ITerm: follow the connections

The screenshot displays the HOTPOT - routed Ariane RISC-V - ariane PCB layout software. The main window shows a complex routing design with various layers and nets. A specific net is highlighted in blue, and a pin is highlighted in yellow. The Inspector panel on the right shows the properties of the selected pin.

Connected net

Library pin

A2 pin

Inspector

Name	Value
Type	ITerm
Name	ex_stage_iv_05454_A2
Instance	ex_stage_iv_05454_
Net	ex_stage_iv_00031_
ModNet	<none>
Special	False
MTerm	NAND2_X1/A2
Access Points	3 items
BBox	(394.31, 774.9), (394.435, 775.075)
BBox Width, Height	(0.125, 0.175)

Added 1 391.9175, 771.602

Net: inspect its endpoints

File View Tools Windows Options Help

Fit Find Inspect Timing

Display Control

Layers

- Other
 - metal1
 - via1
 - metal2
 - via2
 - metal3
 - via3
 - metal4
 - via4
 - metal5
 - via5
 - metal6
 - via6
 - metal7
 - via7
 - metal8
 - via8
 - metal9
 - via9
 - metal10
- Nets
- Instances
- Blockages
- Rulers
- Rows
- Tracks
- Shape Types
- Misc
- Heat Maps
- Timing Path

Click an endpoint

Inspector

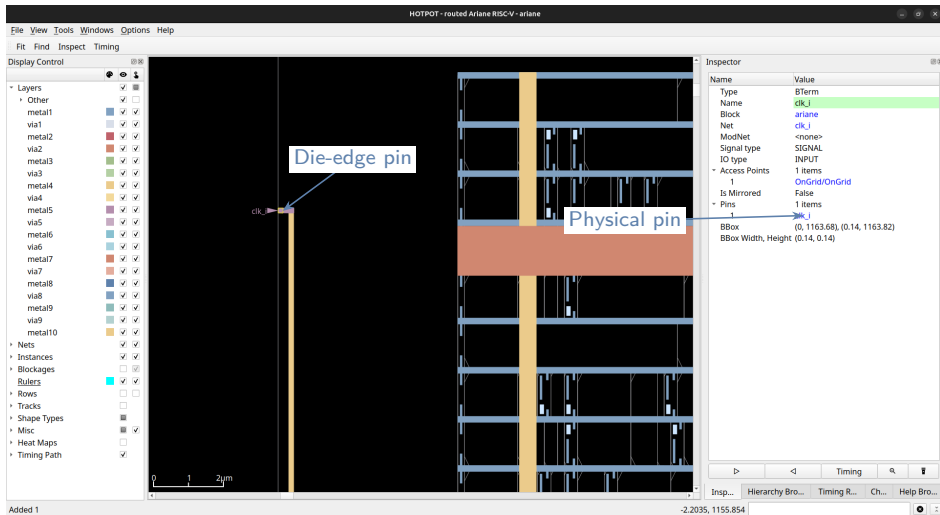
Name	Value
Type	Net
Name	ex_stage_I/_00031_
Block	ariane
Signal type	SIGNAL
Source type	NONE
Wire type	ROUTED
Special	False
Dont Touch	False
ITerms	4 Items
1	ex_stage_I/_05453/_ZN
2	ex_stage_I/_05454/_A2
3	ex_stage_I/_08099/_A1
4	ex_stage_I/_08166/_A2
BTerms	0 Items
Wire	ex_stage_I/_00031_
BBox	(388.745, 770.15), (394.435, 775.075)
BBox Width, Height	(5.69, 4.925)

Timing Show Route Guides

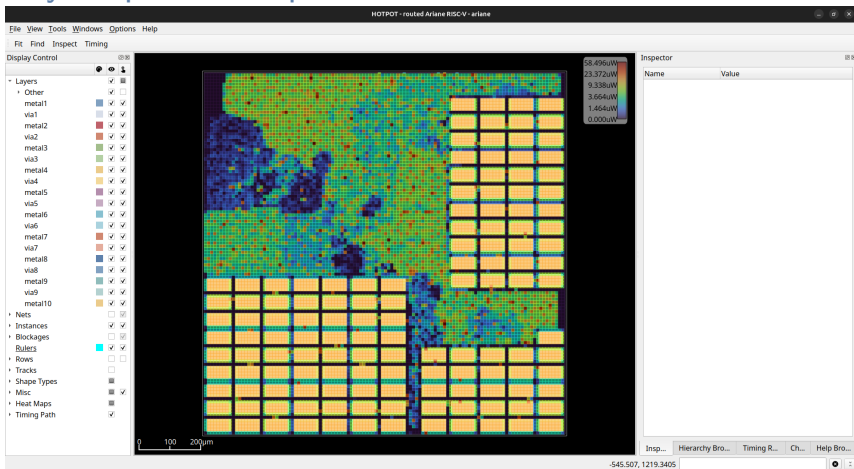
Insp... Hierarchy Bro... Timing R... Ch... Help Bro...

Added 1 386.7495, 766.744

BTerm: a top-level port

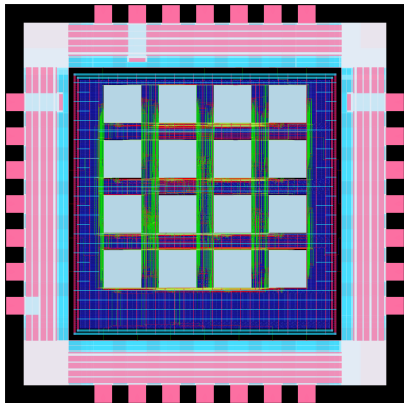


Power density: explore the spatial variation

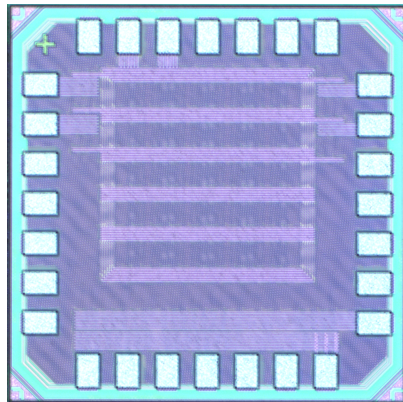


Estimated: 10 ns clock · 0.1 input transitions/cycle · 50% duty
Synthetic RAM models; no workload trace or thermal simulation

Frida: floorplan and fabricated silicon



IHP130 layout



Fabricated TSMC65 die

Similar architecture; different processes and geometries.

The command reference

Database
GUI
Web Viewer
Synthesis
Partition Manager
Restructure
Floorplan Initialization
Pin Placement
Design for Test
Chip-level Connections
Macro Placement
Tapcell Insertion
PDN Generation
Global Placement
Static Timing Analyzer
(external)
Parasitics Estimation
Gate Resizing
Clock gating
Detailed Placement



Options

Switch Name	Description
<code>-core_area</code>	Core area coordinates in microns. Either a rectangle as <code>{llx lly urx ury}</code> , or a rectilinear polygon as a list of vertex coordinates <code>{x1 y1 x2 y2 ...}</code> . See Rectilinear Floorplans .
<code>-core_space</code>	Space around the core in microns. Allowed values are either one value for all margins or a set of four values, one for each margin. The order of the four values are: <code>{bottom top left right}</code> .
<code>-die_area</code>	Die area coordinates in microns. Either a rectangle as <code>{llx lly urx ury}</code> , or a rectilinear polygon as a list of vertex coordinates <code>{x1 y1 x2 y2 ...}</code> . See Rectilinear Floorplans .
<code>-site</code>	Site name.
<code>-utilization</code>	Percentage utilization. Allowed values are <code>double</code> in the range <code>{0-100}</code> .
<code>[-additional_sites]</code>	Tcl list of sites to make rows for (e.g. <code>{SITEXX SITEYY}</code>)

Start the workshop script

```
export OPENROAD_ROOT=/path/to/OpenROAD  
openroad -gui flow.tcl
```

Run from the public repository root.

Read the physical library

```
read_lef $tech/Nangate45_tech.lef
read_lef $tech/Nangate45_stdcell.lef
Technology    Cell masters
```

Read the timing library

```
read_liberty $tech/Nangate45_typ.lib  
1.1 V      25 °C
```

Read and link the netlist

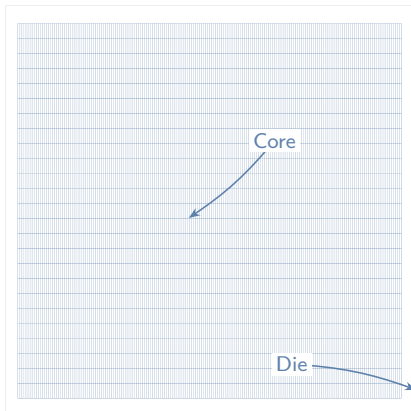
```
read_verilog $netlist
```

```
link_design gcd
```

Top module: gcd

Floorplan comes next

Initialize the floorplan



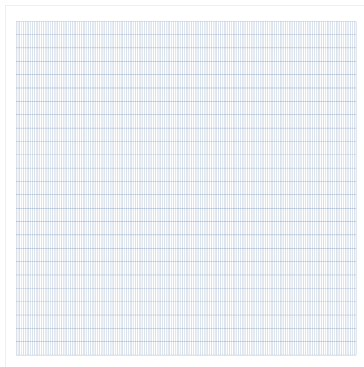
```
initialize_floorplan
  -site <site>
  -utilization <percent>
  -core_space <micrometres>
  [-aspect_ratio <ratio>]
```

Default: ratio 1.0

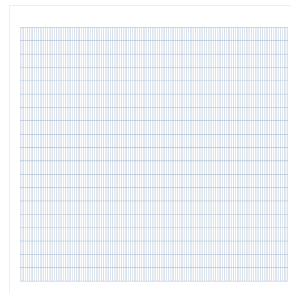
Shown: 40%, margin 1 μm

[argument] optional
<value> placeholder

Utilization sets the available area



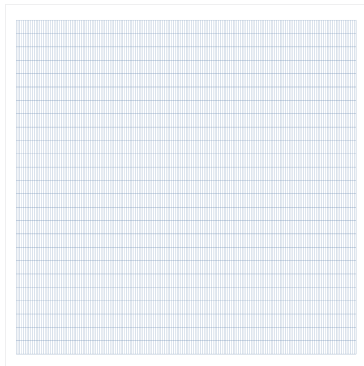
-utilization 40



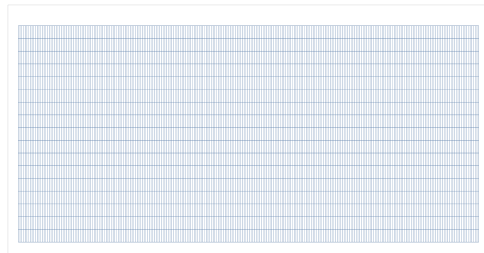
-utilization 65

$$A_{\text{core}} \approx A_{\text{cells}} / (u/100)$$

Choose the shape



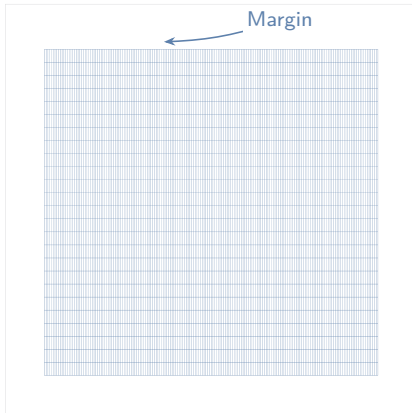
`-aspect_ratio 1`



`-aspect_ratio 0.5`

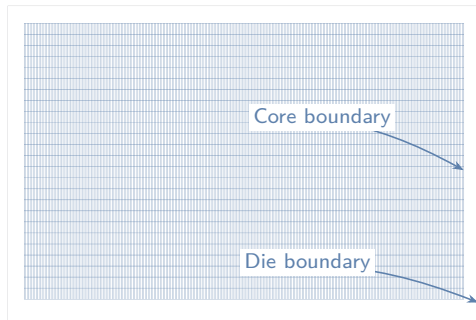
$$r = H/W \quad W = \sqrt{A/r} \quad H = \sqrt{Ar}$$

Reserve the core margin



```
initialize_floorplan  
-site $site  
-utilization 40  
-core_space 4
```

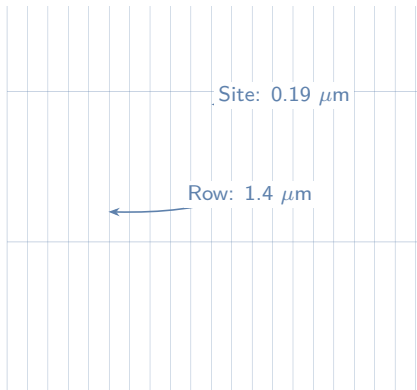
Use a prescribed footprint



```
initialize_floorplan
-site $site
-die_area {0 0 60 40}
-core_area {2 2 58 38}
```

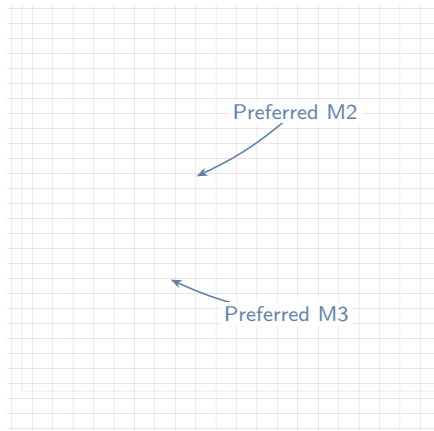
Coordinates in μm

Rows and sites



```
initialize_floorplan  
-site $site  
-utilization 40  
-core_space 1
```

Create routing tracks



`make_tracks`

Pitch and offset from technology LEF

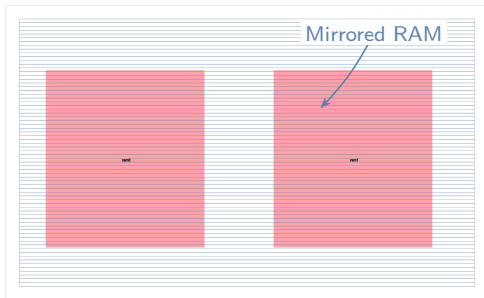
A separate macro example



```
openroad -gui <macros.tcl>
```

Fresh session: two RAM abstracts + GCD

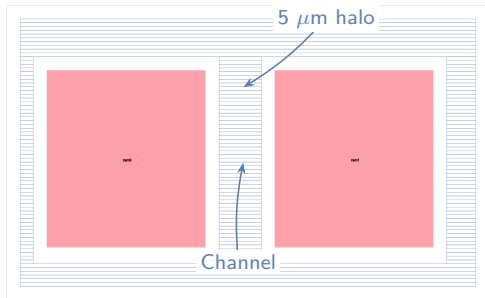
Place and orient a macro



```
place_macro  
  -macro_name ram1  
  -location {100 20}  
  [-orientation MY]
```

Default orientation: R0

Leave access around the macros



```
cut_rows  
[-halo_width_x 5]  
[-halo_width_y 5]
```

Automatic macro placement



`rtl_macro_placer`

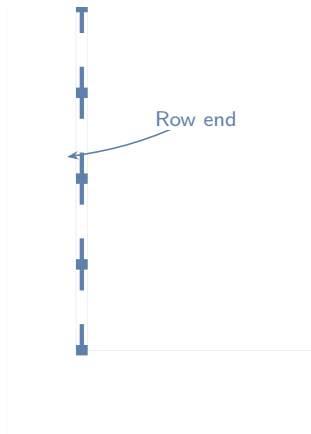
`macros_auto.tcl`: 5 μ m halos

Hierarchical clustering

Simulated annealing

Wirelength + outline + packing

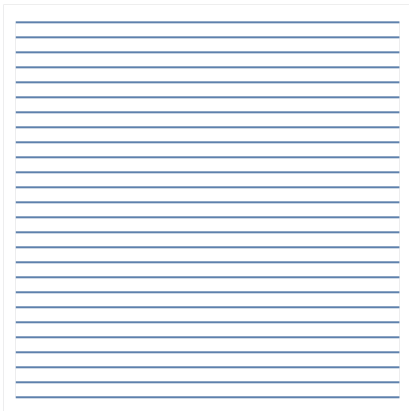
Insert taps and endcaps



```
tapcell  
  [-distance 120]  
  [-tapcell_master TAPCELL_X1]  
  [-endcap_master TAPCELL_X1]
```

GCD branch

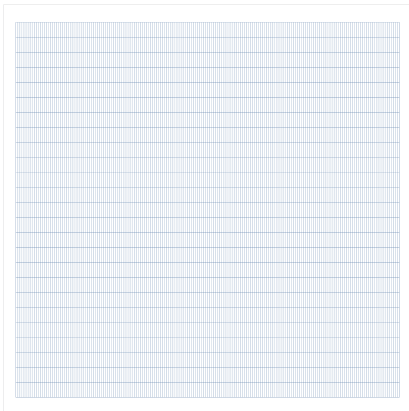
Connect the supply pins



```
add_global_connection  
-net VDD  
-pin_pattern {^VDD$}  
[-power]
```

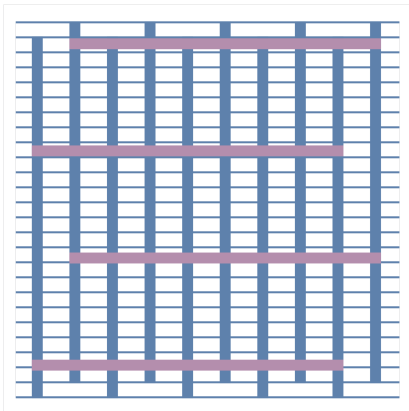
Repeat for VSS with -ground

Define the power domain



```
set_voltage_domain  
-name CORE  
-power VDD  
-ground VSS
```

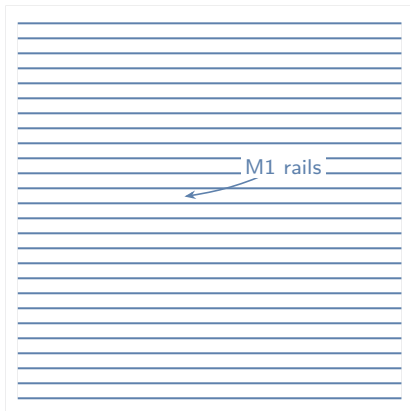
Define the grid



```
define_pdn_grid  
  [-name core_grid]  
  [-voltage_domains CORE]
```

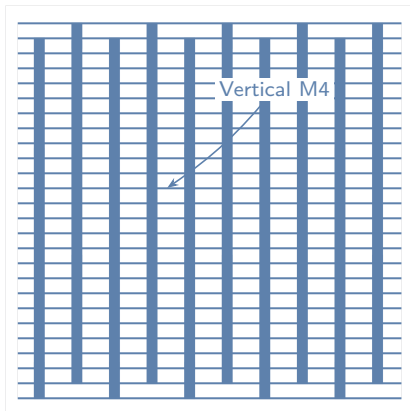
Grid policy; geometry at pdngen

Follow the row power rails



```
add_pdn_stripe  
[-grid core_grid]  
-layer metal1  
[-followpins]
```

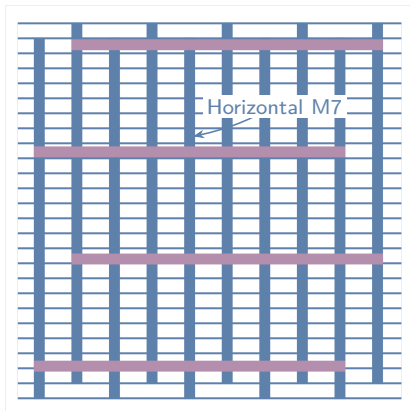
Choose strap width and pitch



```
add_pdn_stripe  
  [-grid core_grid]  
  -layer metal4  
  [-width 1]  
  [-pitch 7]  
  [-offset 2]
```

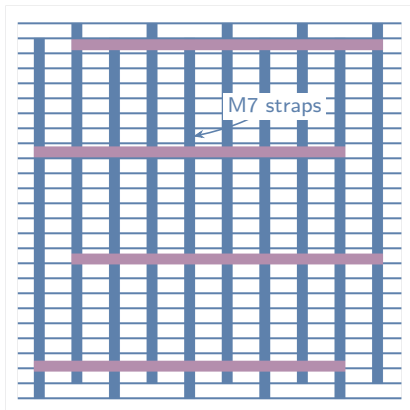
Pitch repeats a VDD/VSS pair

Add a crossing strap layer



```
add_pdn_stripe
-layer metal7
[-width 1]
[-pitch 20]
[-offset 3]
[-number_of_straps 2]
```

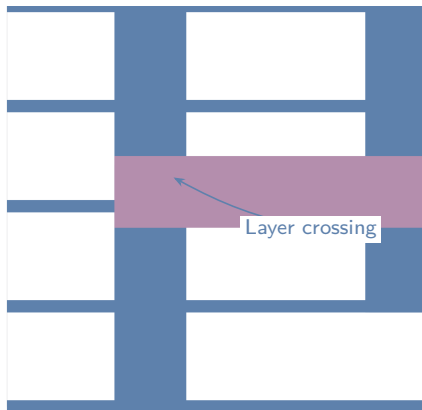
Connect the power layers



```
add_pdn_connect  
  [-grid core_grid]  
  -layers {metal1 metal4}
```

Also connect M4 to M7

Generate the power grid



pdngen

Rule-based straps and via insertion

When do we place the pins?

Pins first

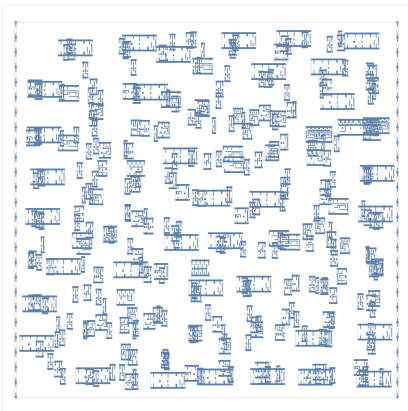
Pin constraints → pin locations → cell placement

Wirelength-driven pin locations

Rough cells → optimize pins → refine cells

Unplaced cell endpoints are modeled at the die center.

Optional: rough cells before pin optimization



```
global_placement  
[-density 0.80]  
[-pad_left 2]  
[-pad_right 2]  
[-skip_io]
```

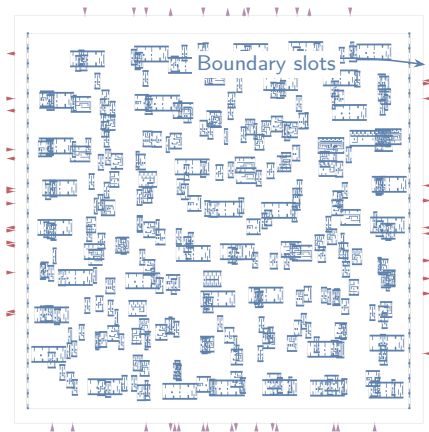
Rough global placement; detailed placement comes later

Smooth wirelength + electrostatic density

Nesterov optimization

Defaults: density 0.7; padding 0

Automatic pin placement



```
place_pins
```

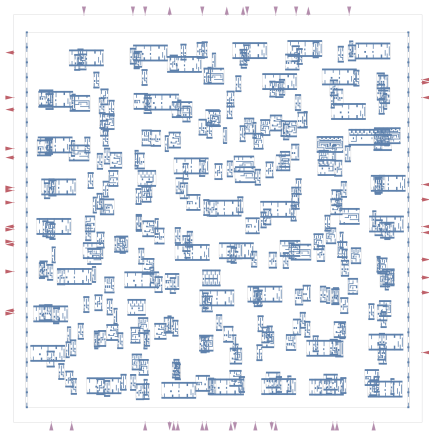
```
-hor_layers metal3
```

```
-ver_layers metal2
```

Fixed cell locations + track slots

Unplaced cells: modeled at die center

The pin-assignment objective



$$c_{ps} = \text{HPWL}(S_p \cup \{s\})$$

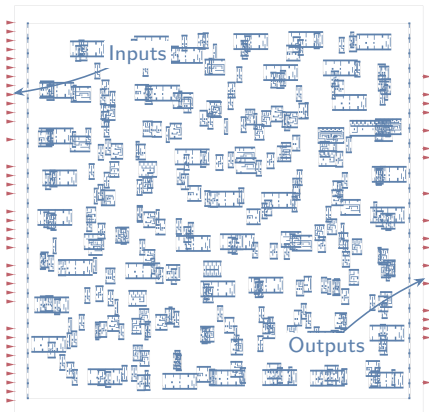
S_p : fixed cell-pin locations

Hungarian matching within slot sections

Optional simulated annealing

Geometric cost; no routed paths

Restrict pins to an edge

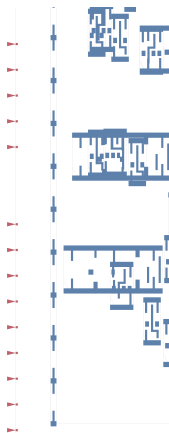


```
set_io_pin_constraint  
  [-direction input]  
  [-region left:*
```

Outputs: -region right:*

Then rerun place_pins

Control the minimum separation



```
place_pins
```

```
-hor_layers metal3
```

```
-ver_layers metal2
```

```
[-min_distance 0.8]
```

0.8 μ m minimum; track snapped

-min_distance_in_tracks: track count

Default separation: two routing tracks

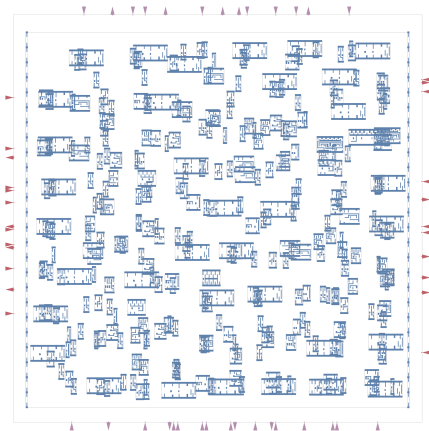
Keep a bus together



```
set_io_pin_constraint  
[-pin_names {resp_msg[*]}]  
[-region right:~]  
[-group]  
[-order]
```

Then rerun place_pins

Try the annealing solver

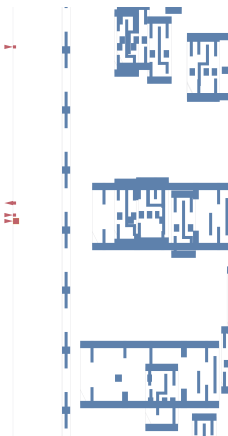


```
place_pins  
  -hor_layers metal3  
  -ver_layers metal2  
  [-annealing]
```

Same HPWL objective

Try edge + bus constraints

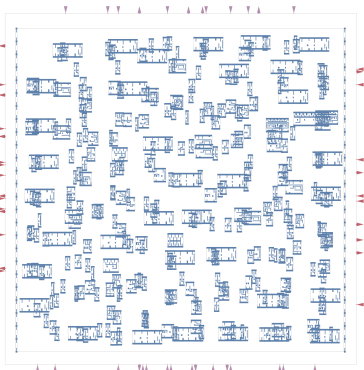
Fix one pin first



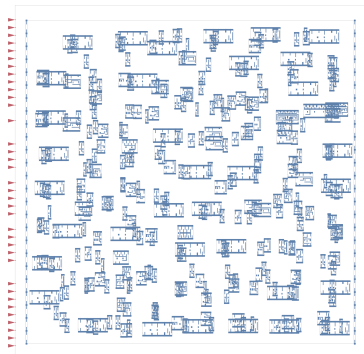
```
place_pin
  -pin_name clk
  -layer metal3
  -location {0 10}
  [-force_to_die_boundary]
```

Then place_pins for the remaining ports

Measure the cost of constraints



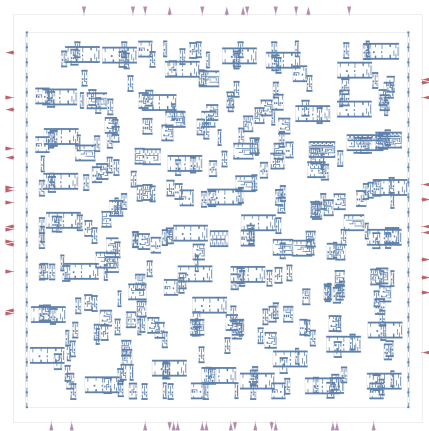
Automatic: 536.82 μm



Constrained: 1126.27 μm

Same fixed cells; constraints: opposite edges + 0.8 μm spacing

Refine cells with the pins included



```
global_placement  
  [-routability_driven]  
  [-density 0.80]  
  [-pad_left 2]  
  [-pad_right 2]
```

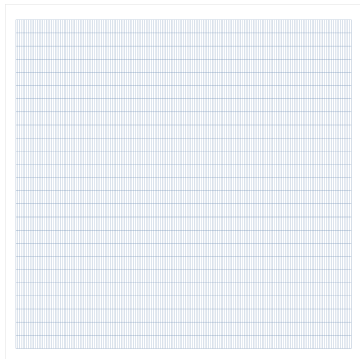
Pins now contribute to the placement objective

RUDY congestion estimate + cell inflation

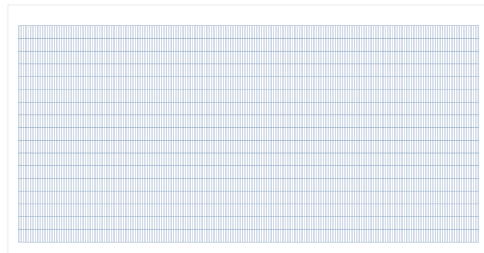
-routability_use_grt: FastRoute feedback

Defaults: density 0.7; padding 0

Exercise: compare two floorplans



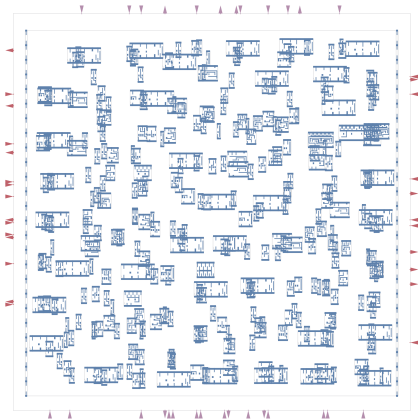
$$u = 40\%, H/W = 1$$



$$u = 40\%, H/W = 0.5$$

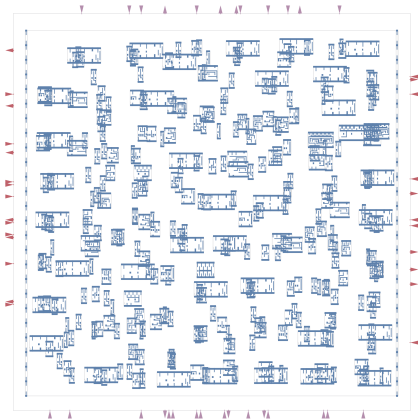
Compare I/O HPWL and pin constraints after placement

Save a half-finished design: ODB



```
write_db  
    partial.odb  
Physical database checkpoint
```

Resume the ODB in a fresh session



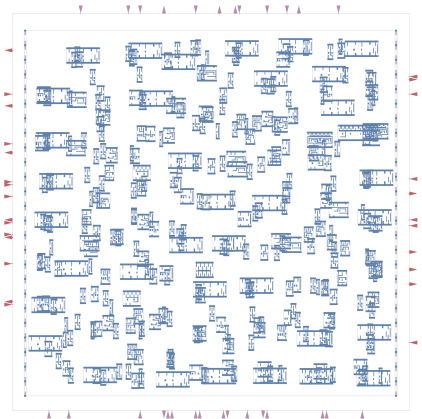
```
read_db
```

```
    partial.odb
```

Technology + cell masters are included

Timing: also read Liberty + SDC

Save a half-finished design: DEF

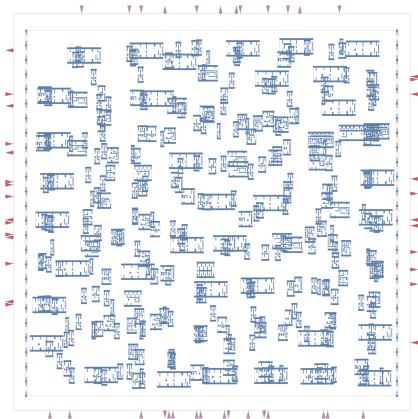


```
write_def
```

```
    partial.def
```

Current rows, pins and placement; routes if present

Read the DEF in a fresh session



```
read_lef $tech/Nangate45_tech.lef
read_lef $tech/Nangate45_stdcell.lef
```

```
read_def
    partial.def
```

LEFs supply technology + cell masters
Timing: also read Liberty + SDC

Open the workshop checkpoints

```
openroad -gui examples/floorplanning/read_odb.tcl  
openroad -gui examples/floorplanning/read_def.tcl  
Separate sessions; run from slides/.
```

```
cd examples/ariane  
make gui
```

Ariane: included completed checkpoint + standalone OpenROAD.